<!--Estudio IMIBIC-->

R para Todos: El poder para explotar tus Datos {

<Por="Adrián Santiago Ortiz"/>

}



Equipo {



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UCAIB BIOINFORMÁTICA Y BIOESTADÍSTICA (IMIBIC)

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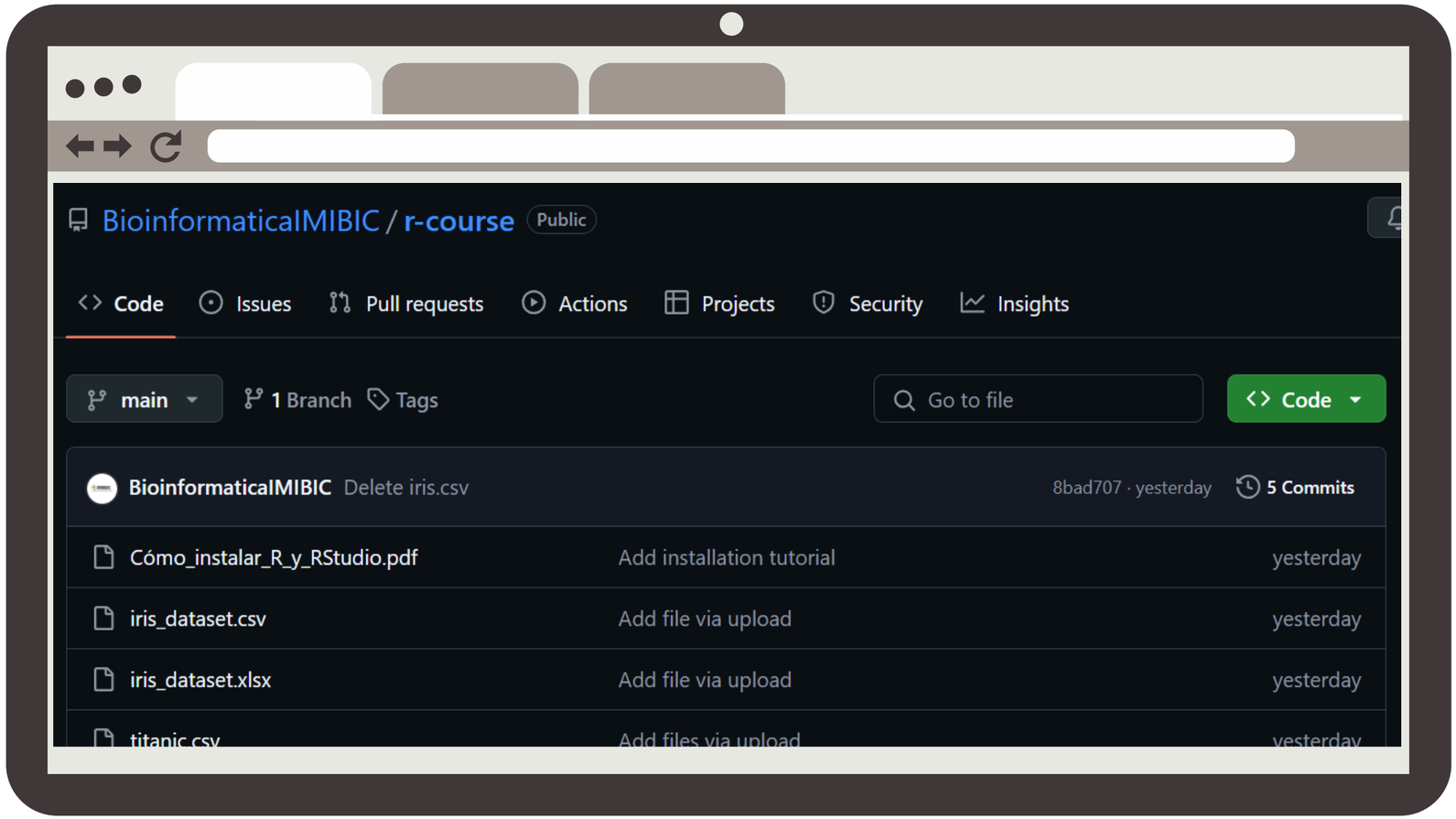
ALBERTO VELA

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}

Material disponible



https://github.com/BioinformaticaIMIBIC/r-course

Contenidos

01 Introducción a R y Rstudio

02 Tipos de datos

03 Operadores en R

04 Tipos de objetos y operaciones

05 Estructuras de control y flujo de ejecución

06 Creación de funciones

07 Visualización de resultados en gráficos

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02 Tipos de datos

03 Operadores en R

04 Tipos de objetos y operaciones

Sesión 1

05 Estructuras de control y flujo de ejecución

06 Creación de funciones

07 Visualización de resultados en gráficos

Contenidos

01 Introducción a R y Rstudio

02

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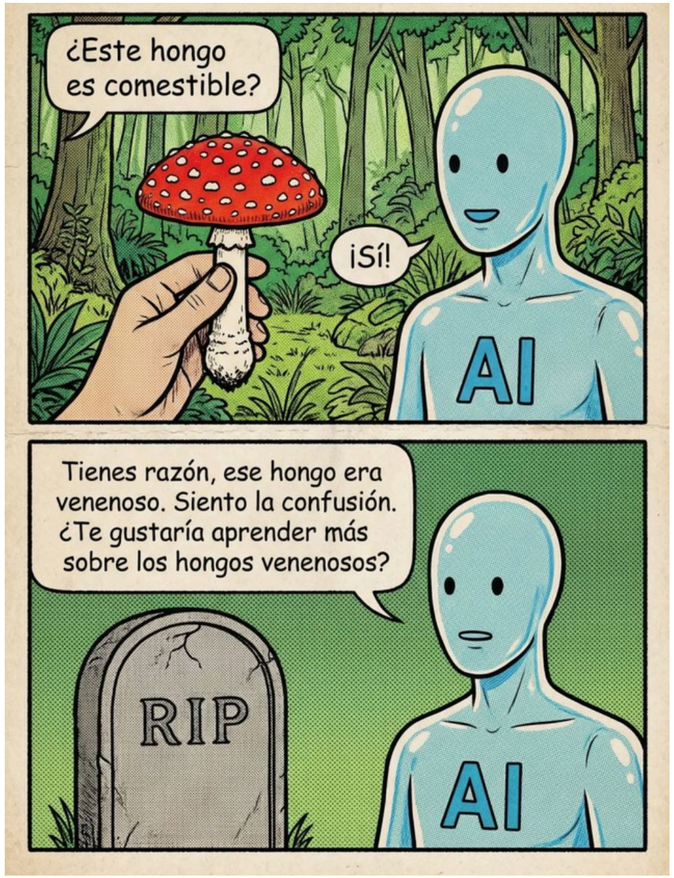
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Contenidos

01 Introducción a R y Rstudio

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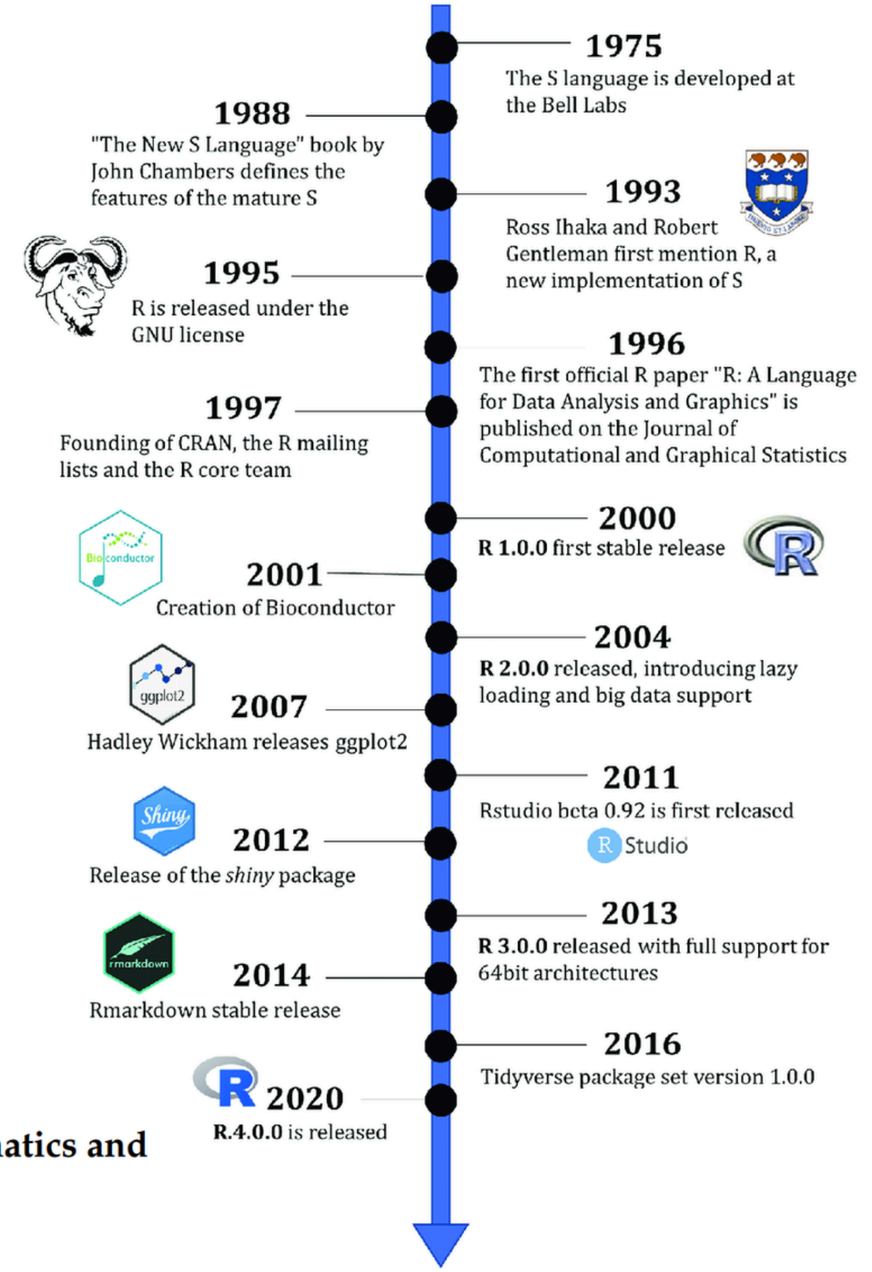
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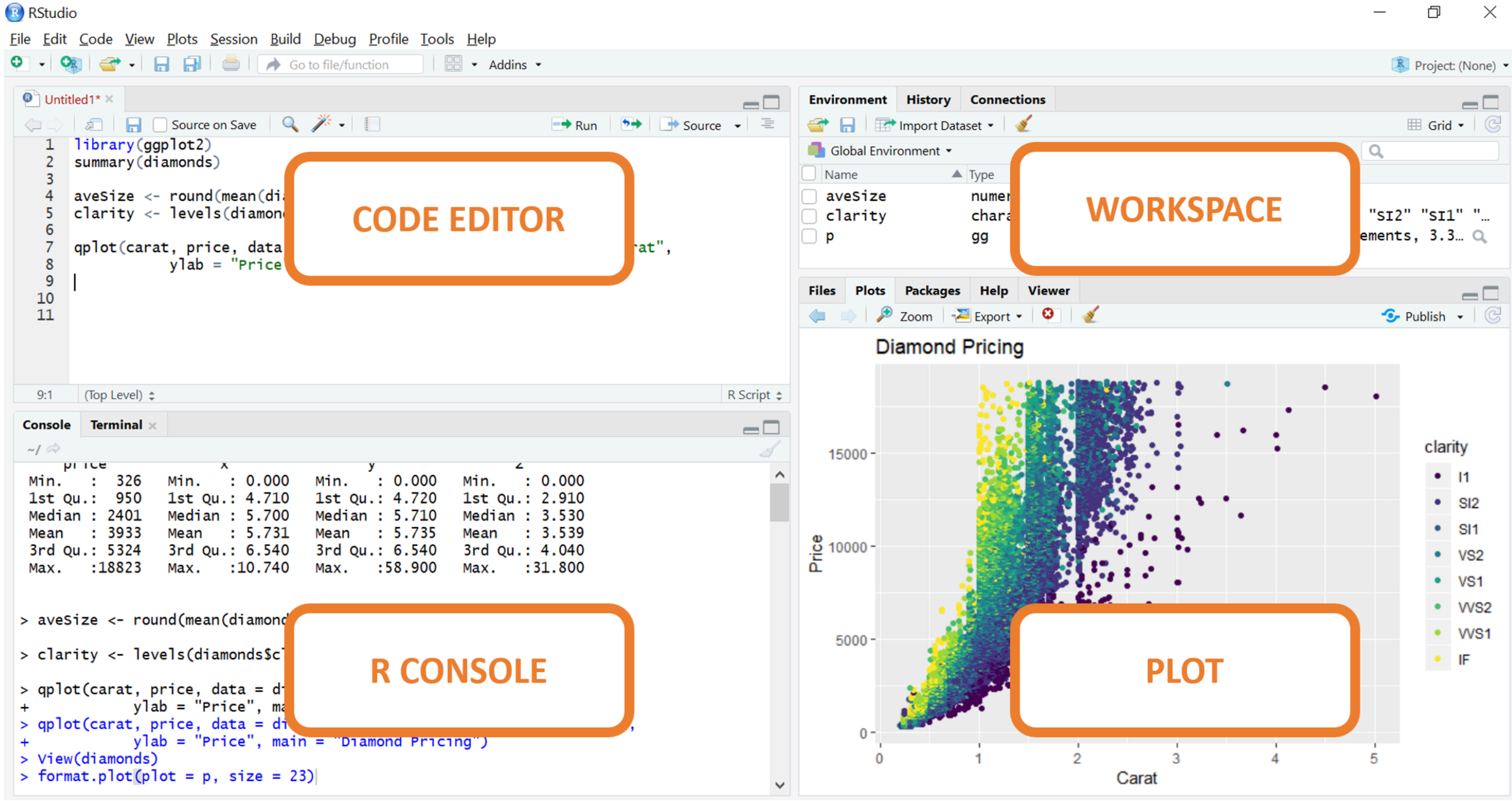


Review

The R Language: An Engine for Bioinformatics and Data Science

Federico M. Giorgi [1,*], Carmine Ceraolo [1,2] and Daniele Mercatelli [1]

Un vistazo a RStudio®

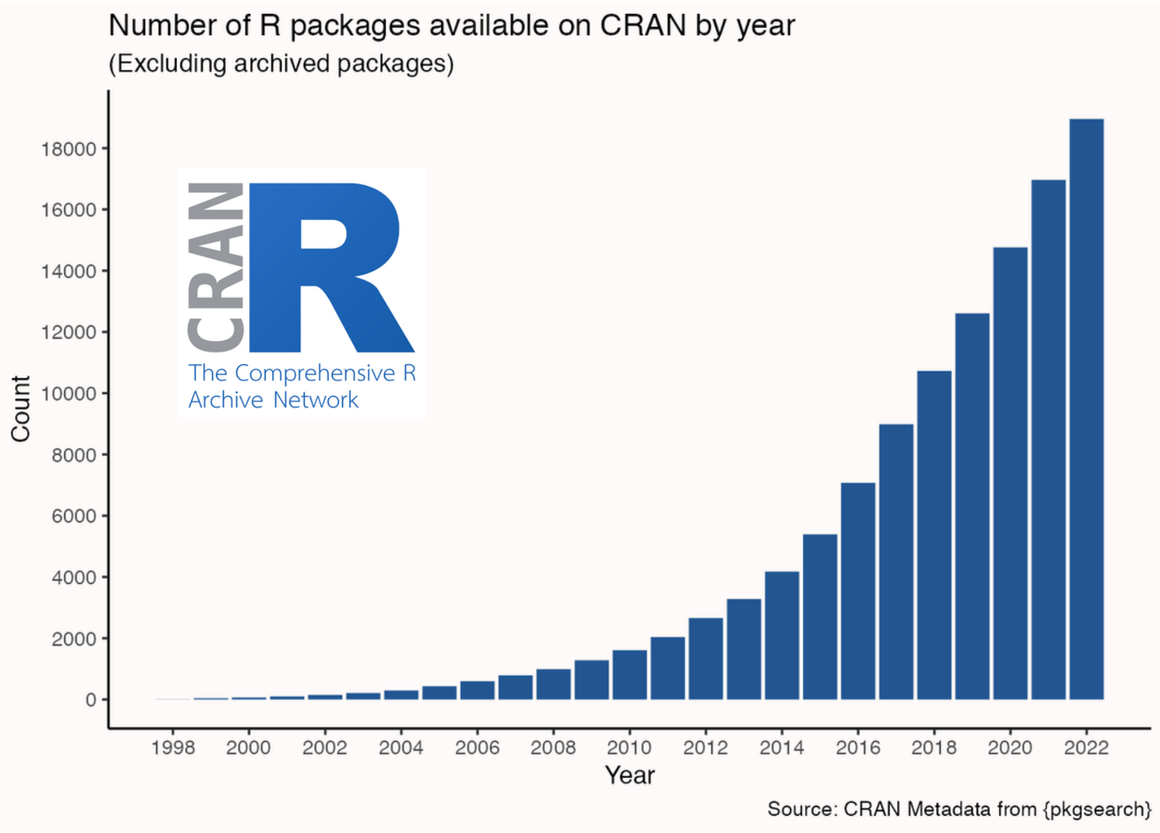


Librerías e importación de datos

```
install.packages("package1")
library(package1)
```

Bioconductor
OPEN SOURCE SOFTWARE FOR BIOINFORMATICS





Librerías e importación de datos

```
read.csv("file.csv", ...)
```




```
read.table("file.txt", ...)
```








```
install.packages("readxl")
library(readxl)

read_excel("file.xlsx", ...)
```

Contenidos

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Operadores aritméticos



Operador	Descripción	Ejemplo	Resultado
+	Suma	23 + 16	
–	Resta	12 - 7	
*	Multiplicación	8 * 6	
/	División	5 / 2	
^ ó **	Potenciación	10 ** 3	
%%	Módulo	5 %% 3	
%/%	División entera	8 %/% 6	

ORDEN	OPERADORES
1	^
2	* /
3	+ –
4	< > <= >= == !=
5	!
6	&
7	|
8	<–

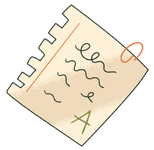
Operadores aritméticos



Operador	Descripción	Ejemplo	Resultado
+	Suma	23 + 16	39
–	Resta	12 - 7	5
*	Multiplicación	8 * 6	48
/	División	5 / 2	2.5
^ ó **	Potenciación	10 ** 3	1000
%%	Módulo	5 %% 3	2
%/%	División entera	8 %/% 6	1

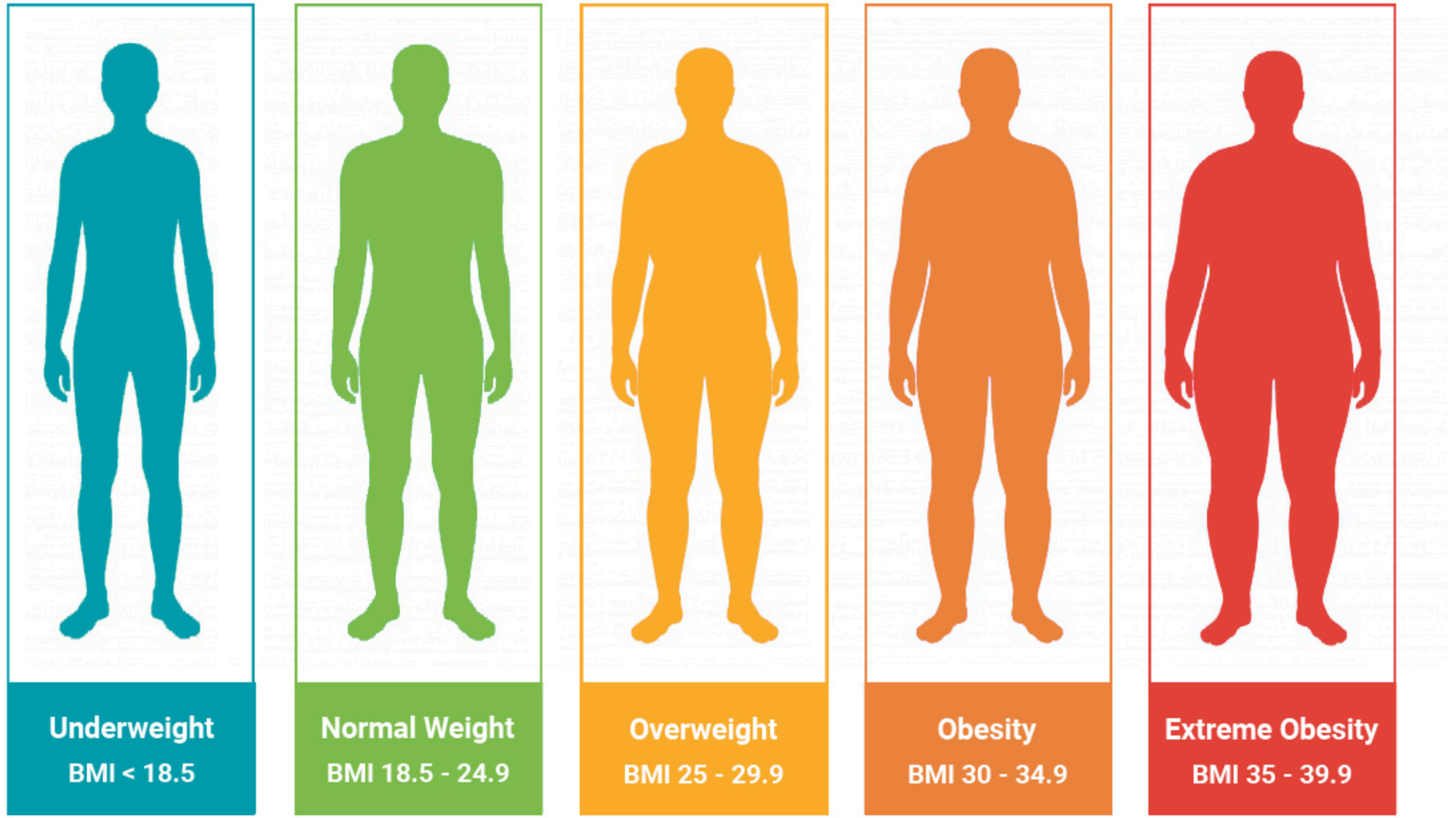
ORDEN	OPERADORES
1	^
2	* /
3	+ -
4	< > <= >= == !=
5	!
6	&
7	|
8	<-

Ejercicios

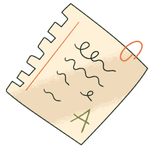


Calcula el IMC de un paciente de 68 kg y 1.70 m

$$\text{BMI} = \frac{\text{Weight (in kilograms)}}{\text{Height}^2 \text{ (in meters)}}$$



Ejercicios



Calcula el nº de células bacterianas en una población inicial de 1e6 con una tasa de crecimiento de 0.3 tras 5 horas

Exponential Growth

$$P = P_0 e^{rt}$$

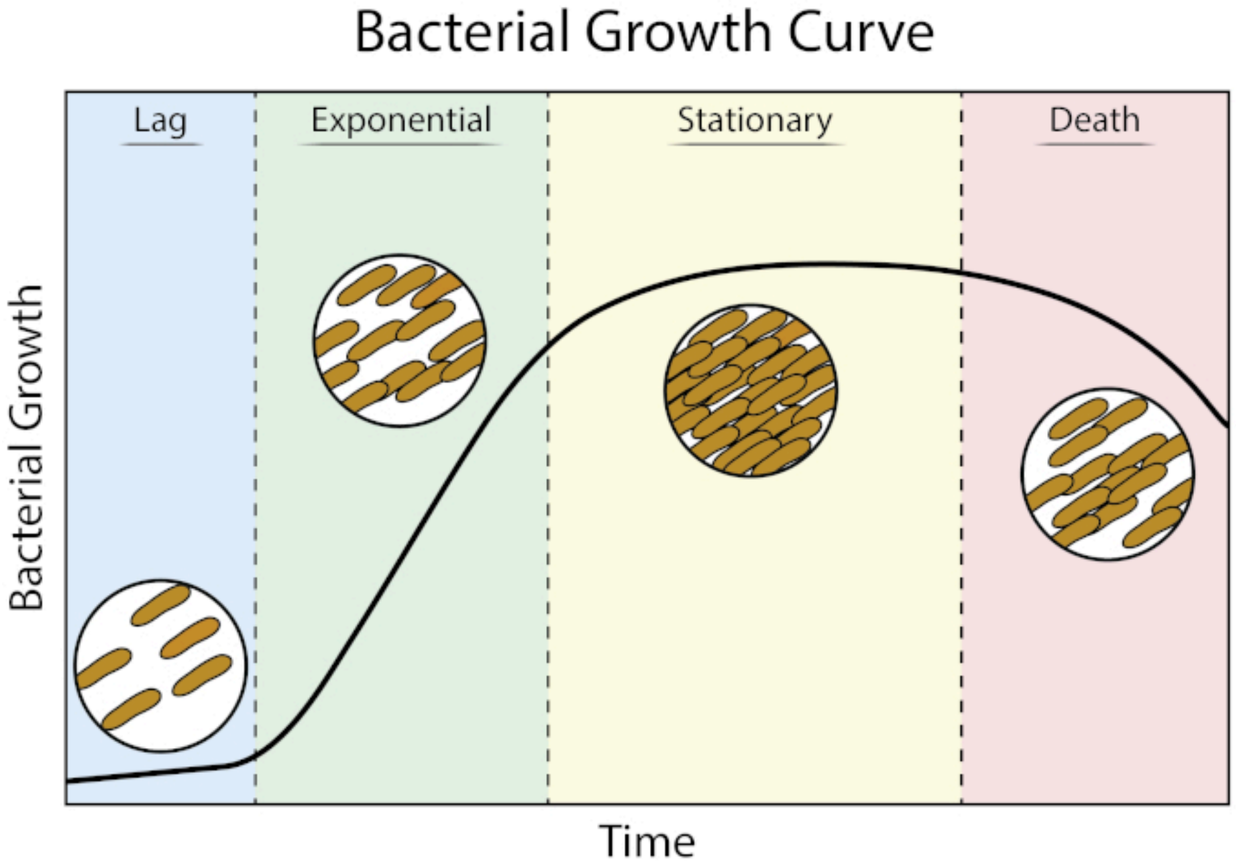
P = total population after time t

P_0 = starting population

r = % rate of growth

t = time

e = Euler's number



Operadores relacionales y lógicos

TRUE or FALSE?

```
vector <- c(1,3,5,7,9)
```

Operador	Descripción	Ejemplo	Resultado
`< y >`	menor/mayor a	vector > 3	
`<= y >=`	menor/mayor o igual a	vector >= 3	
`==`	exactamente igual a	vector == 3	
`!=`	no es igual a	vector != 3	
`x|y`	x OR y	vector [vector >= 5 | vector < 3]	
`x & y`	x AND y	vector [vector >=3 & vector < 7]	
`x %in% y`	x IN y	vector [vector %in% c(3,7)]	

Operadores relacionales y lógicos

TRUE or FALSE?

```
vector <- c(1,3,5,7,9)
```

Operador	Descripción	Ejemplo	Resultado
`< y >`	menor/mayor a	vector > 3	FALSE, FALSE, TRUE, TRUE, TRUE
`<= y >=`	menor/mayor o igual a	vector >= 3	FALSE, TRUE, TRUE, TRUE, TRUE,
`==`	exactamente igual a	vector == 3	FALSE, TRUE, FALSE, FALSE, FALSE
`!=`	no es igual a	vector != 3	TRUE, FALSE, TRUE, TRUE, TRUE
`x|y`	x OR y	vector [vector >= 5 | vector < 3]	1, 5, 7, 9
`x & y`	x AND y	vector [vector >=3 & vector < 7]	3, 5
`x %in% y`	x IN y	vector [vector %in% c(3,7)]	3, 7

Contenidos

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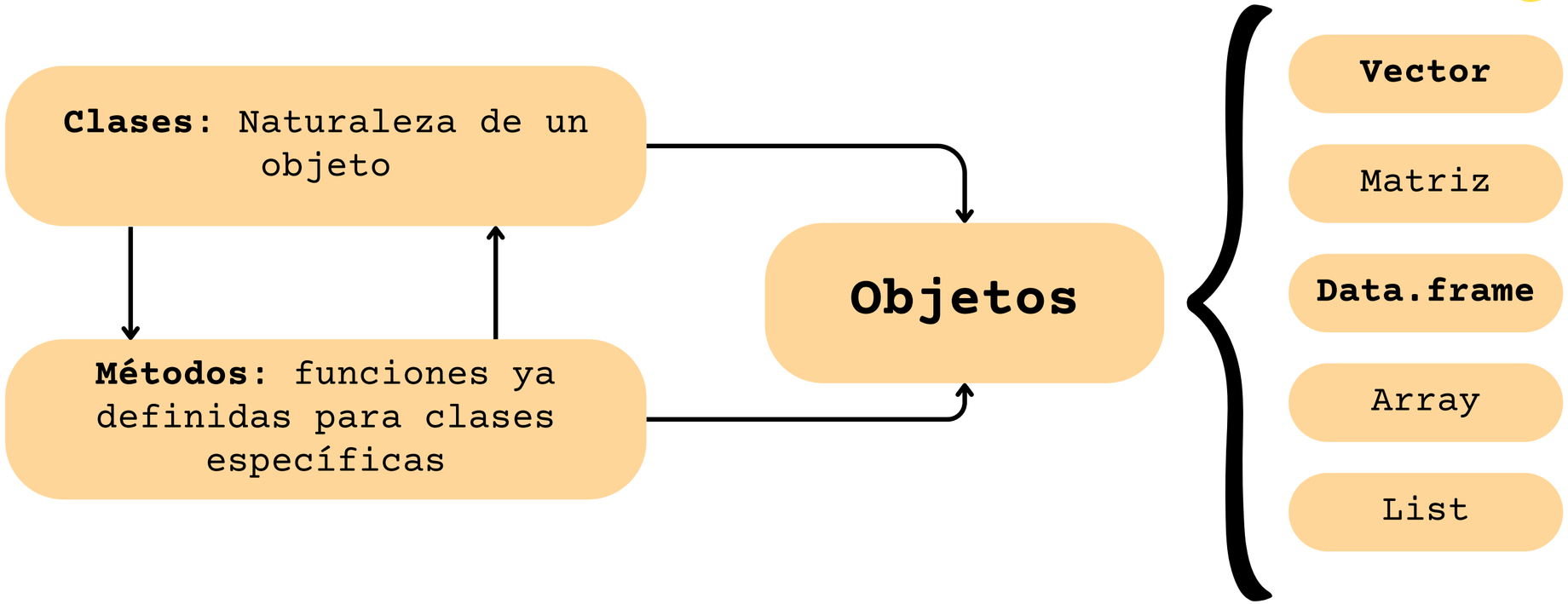
07

Otro vistazo a RStudio®

OOP

Object-**O**riented **P**rogramming

"Everything in R is an object"

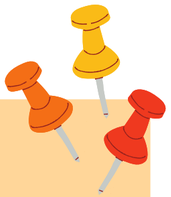


Otro vistazo a RStudio®

	Dimensions	Mode (data "type")	Example
Vector	1 	Identical	`c(10,0.2,34,48,53)`

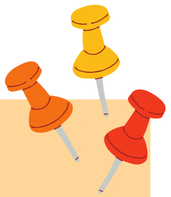
Fuente: https://r.qcbs.ca/workshop01/pres-en/workshop01-pres-en.html#100

Tips para operar con vectores



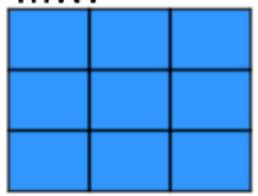
Función	Ejemplo	Resultado
c(a, b, ...)	c(1, 3, 5, 7, 9)	
a:b	1:10	
seq(from, to, by, length.out)	seq(from = 10, to = 50, by = 5)	
rep(x, times, each, length.out)	rep(c(4, 5), times = 3, each = 2)	

Tips para operar con vectores



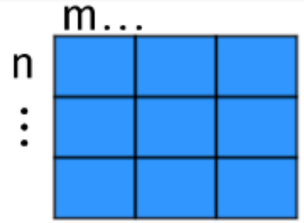
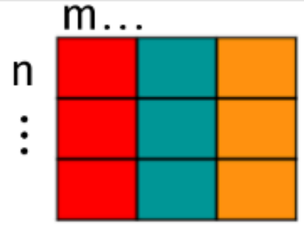
Función	Ejemplo	Resultado
c(a, b, ...)	c(1, 3, 5, 7, 9)	1, 3, 5, 7, 9
a:b	1:10	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
seq(from, to, by, length.out)	seq(from = 10, to = 50, by = 5)	10, 15, 20, 25, 30, 35, 40, 45, 50
rep(x, times, each, length.out)	rep(c(4, 5), times = 3, each = 2)	4, 4, 5, 5, 4, 4, 5, 5, 4, 4, 5, 5

Otro vistazo a RStudio®

	Dimensions	Mode (data "type")	Example
Vector	1 	Identical	`c(10,0.2,34,48,53)`
Matrix	n 	Identical	`matrix(c(1,2,3, 11,12,13), nrow = 2, ncol = 3)`

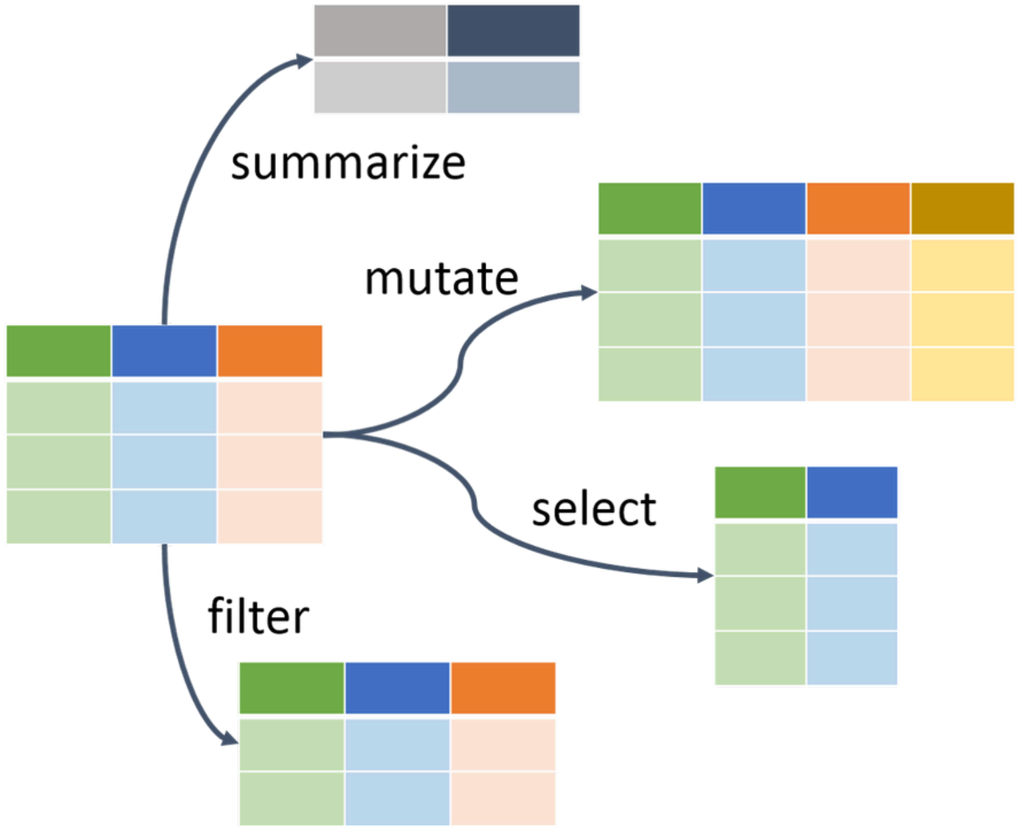
Fuente: https://r.qcbs.ca/workshop01/pres-en/workshop01-pres-en.html#100

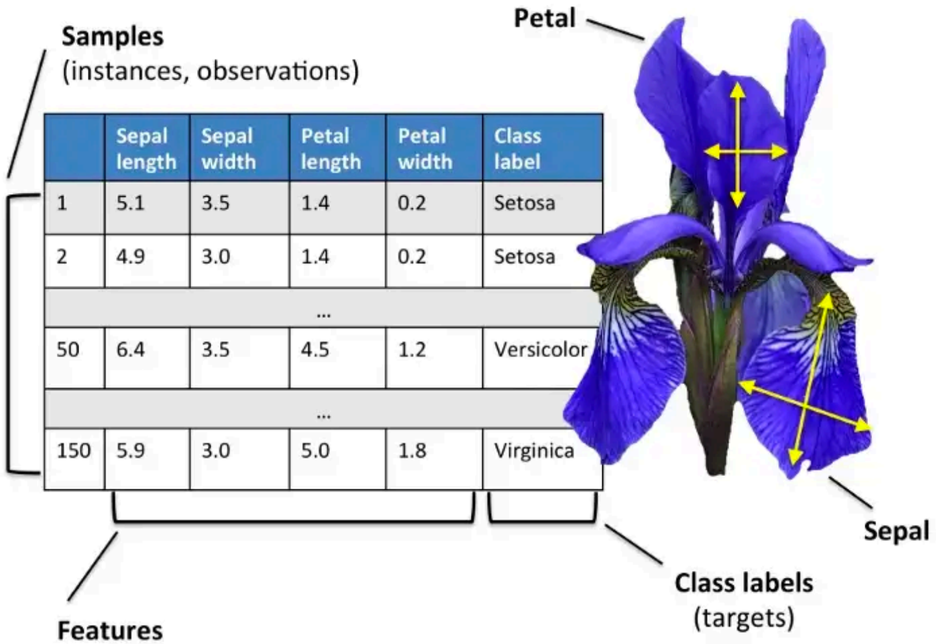
Otro vistazo a RStudio®

	Dimensions	Mode (data "type")	Example
Vector		Identical	`c(10,0.2,34,48,53)`
Matrix		Identical	`matrix(c(1,2,3, 11,12,13), nrow = 2, ncol = 3)`
Data frame		Can be different	`data.frame(x = 1:3, y = 5:7)`

Fuente: https://r.qcbs.ca/workshop01/pres-en/workshop01-pres-en.html#100

Paquete





	Sepal length	Sepal width	Petal length	Petal width	Class label
1	5.1	3.5	1.4	0.2	Setosa
2	4.9	3.0	1.4	0.2	Setosa
...					
50	6.4	3.5	4.5	1.2	Versicolor
...					
150	5.9	3.0	5.0	1.8	Virginica

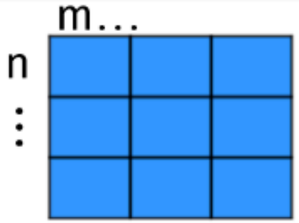
- 150 observaciones del género Iris
- Cuatro características
- Análisis de clasificación, (Regresión Logística, KNN y árboles de decisión)

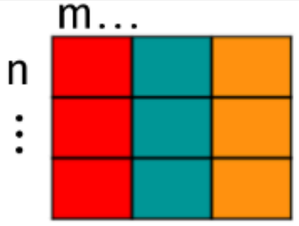
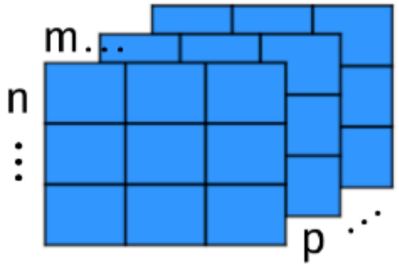


Fuente: https://eminebozkus.medium.com/exploring-the-iris-flower-dataset-4e000bcc266c

Otro vistazo a RStudio®

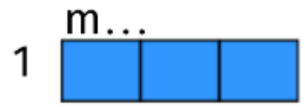
	Dimensions	Mode (data "type")	Example
Vector		Identical	`c(10,0.2,34,48,53)`

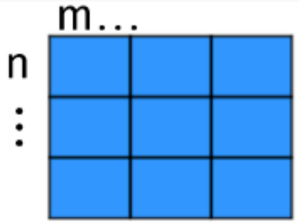
Matrix		Identical	`matrix(c(1,2,3,`
			`11,12,13),`
			`nrow = 2,`
			`ncol = 3)`

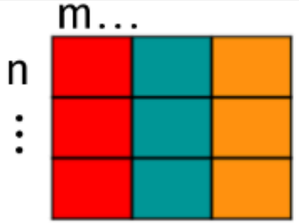
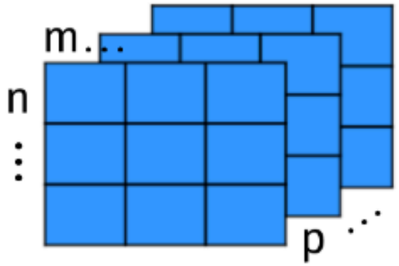
Data frame		Can be different	`data.frame(x = 1:3,` `y = 5:7)`
Array		Identical	`array(data = 1:3,` `dim = c(2,4,2))`

Fuente: https://r.qcbs.ca/workshop01/pres-en/workshop01-pres-en.html#100

Otro vistazo a RStudio®

	Dimensions	Mode (data "type")	Example
Vector		Identical	`c(10,0.2,34,48,53)`

Matrix		Identical	`matrix(c(1,2,3,`
			`11,12,13),`
			`nrow = 2,`
			`ncol = 3)`

Data frame		Can be different	`data.frame(x = 1:3,` `y = 5:7)`
Array		Identical	`array(data = 1:3,` `dim = c(2,4,2))`
List	Vector Matrix Data frame Array	Can be different	`list(x = cars[,1],` `y = cars[,2])`

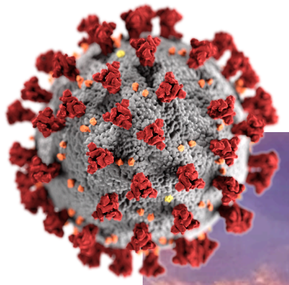
Fuente: https://r.qcbs.ca/workshop01/pres-en/workshop01-pres-en.html#100

Para practicar...

Deidentified Results of COVID-19 testing at the Children's Hospital of Pennsylvania (CHOP) in 2020.

Data frame: covid_testing

- 15524 observaciones
- 17 variables



Children's Hospital of Philadelphia
RESEARCH INSTITUTE




```
install.packages("medicaldata")
library(medicaldata)

data(covid_testing)
```

<!--Estudio IMIBIC-->

Enhorabuena {

<"Has superado la primera sesión"/>

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